

ML Week 5

Number of participants: 89



1. Logistic regression

4 correct answers
out of 16 respondents

is a non-linear
classification
method.



9 votes



can overfit the data.



10 votes

always has a
closed-form
solution.



2 votes



is fitted by
maximum
likelihood
estimation.



12 votes



2. LDA and QDA

1 correct answer
out of 16 respondents



QDA is more flexible than LDA.



12 votes

QDA always has a smaller test missclassification rate than LDA.



3 votes



LDA and QDA can both predict the class probabilities of a new test observation.



12 votes



LDA and QDA cannot handle qualitative predictors.



2 votes



3. LDA versus logistic regression

3 correct answers
out of 14 respondents

Logistic regression finds the linear separation with the lowest misclassification error.



4 votes

For the same data set, both LDA and log. regr. give linear but possibly different decision boundaries.



9 votes

With infinitely many samples, both LDA and log. regr. attain the Bayes error rate.



5 votes

Both methods can be used to predict the probability of a class.



8 votes



If you standardize the inputs, then
4. you influence the predictions of
which method(s)?

5 correct answers
out of 14 respondents



k-nearest
neighbors (kNN)



9 votes

Logistic regression



3 votes

LDA



8 votes

QDA



7 votes